Chapter 17: Fabrication and Processing of Engineering Materials

ISSUES TO ADDRESS...

- What are some of the common fabrication techniques for metals?
- What heat treatment procedures are used to improve the mechanical properties of both ferrous and nonferrous alloys?
- How is processing of ceramics different than for metals?
- What are the primary polymer processing methods?

Metal Fabrication

- How do we fabricate metals?
  - Blacksmith - hammer (forged)
  - Cast molten metal into mold

Hot working vs. Cold working

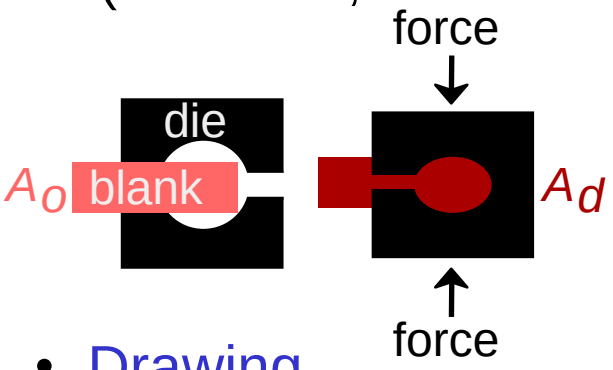
- Deformation temperature high enough for recrystallization
- Large deformations

- Deformation below recrystallization temperature
- Strain hardening occurs
- Small deformations

Metal Fabrication Methods (i)

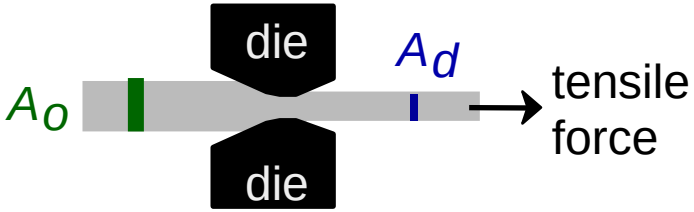
FORMING | CASTING | MISCELLANEOUS

- Forging (Hammering; Stamping)
  (wrenches, crankshafts)



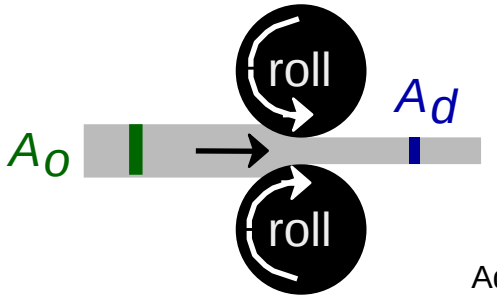
often at elev. *T*

- Drawing
  (rods, wire, tubing)

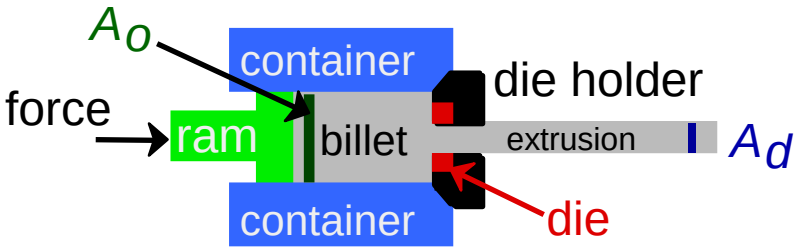


die must be well lubricated & clean

- Rolling (Hot or Cold Rolling)
  (I-beams, rails, sheet & plate)



- Extrusion
  (rods, tubing)



ductile metals, e.g. Cu, Al (hot)

Adapted from Fig. 17.2, *Callister & Rethwisch 9e.*

Forging is a manufacturing process involving the shaping of a metal through hammering or pressing. These compressive forces are delivered with a hammer or die. Forging is often categorized according to the temperature at which it is performed—cold or hot forging.

Hot Forging

Compressive force squeezes the metal between a punch and die at temperature above the recrystallization temperature of the metal until the material conforms to the die's contours.

Cold Forging

Compressive force squeezes the metal between a punch and die at temperatures below the recrystallization temperature until the material conforms to the die's contours.

Rolling process is best defined as the shaping of metals into semi-finished or finished forms by passing between rollers rotating in opposite direction.

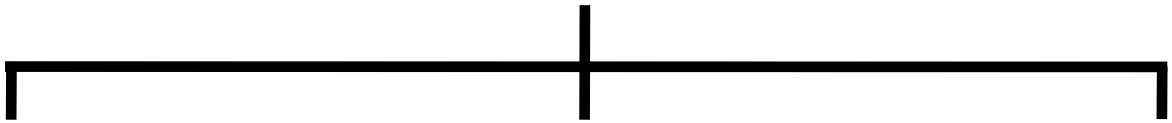
Rolling is the most important and widely used metal forming process because of its lower cost and higher productivity.

Rolling is classified according to the temperature of the metal rolled. If the temperature of the metal is above its recrystallization temperature, then the process is known as **hot rolling**. If the temperature of the metal is below its recrystallization temperature, the process is known as **cold rolling**.

Metal drawing is a metalworking process by which metal is stretched and formed using tensile force. The process is typically performed cold, which is done to ensure accurate tolerances, better grain structure and surface finish, as well as improved material properties.

Extrusion is a process where a material undergoes plastic deformation by the application of a force causing that material to flow through an orifice or die. The material adopts the cross-sectional profile of the die and if the material has suitable properties, that shape is retained in the final extrudate.

Metal Fabrication Methods (ii)



FORMING CASTING MISCELLANEOUS

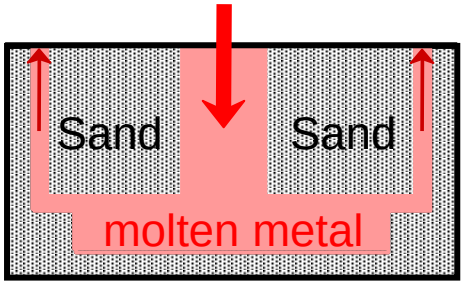
- **Casting**- is a manufacturing process in which a liquid metal is usually poured into a mold, which contains a hollow cavity of the desired shape, and then allowed to solidify.

- The mold is made of 2 parts, the Cope and the Drag.

  - gives good production of shapes
  - weaker products, internal defects
  - good option for brittle materials
  - common and inexpensive.

Types of Casting

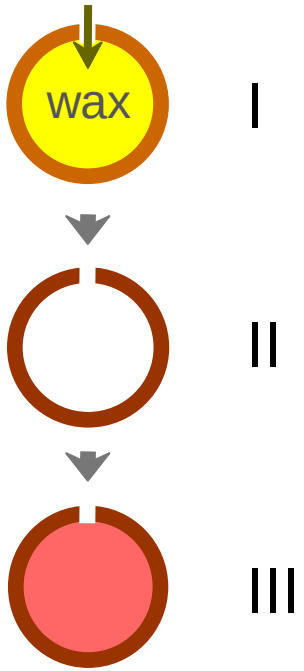
- **Sand Casting**
  (large parts, e.g.,
  auto engine blocks)



- *Sand casting*, is a metal casting process characterized by using sand as the mold material.
- To create mold, pack sand around form (pattern) of desired shape

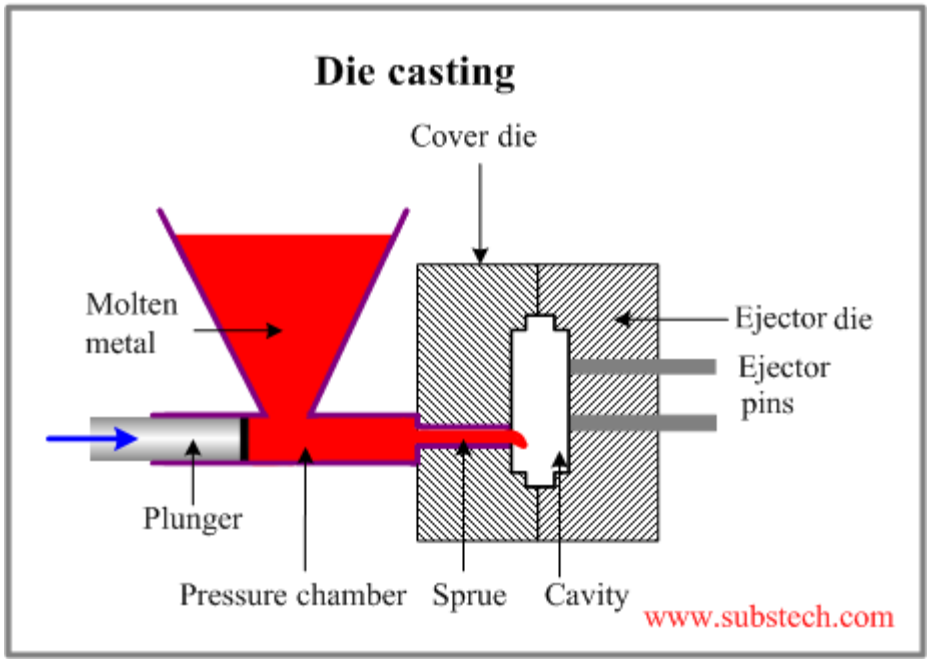
- **Investment casting**, also known as **precision casting** or **lost-wax casting**, is a manufacturing process in which a wax pattern is used to shape a disposable ceramic mold.
(low volume, complex shapes., jewelry, turbine blades)

- Stage I — Mold formed by pouring plaster of paris around wax pattern. Plaster allowed to harden.
- Stage II — Wax is melted and then poured from mold—hollow mold cavity remains.
- Stage III — Molten metal is poured into mold and allowed to solidify.

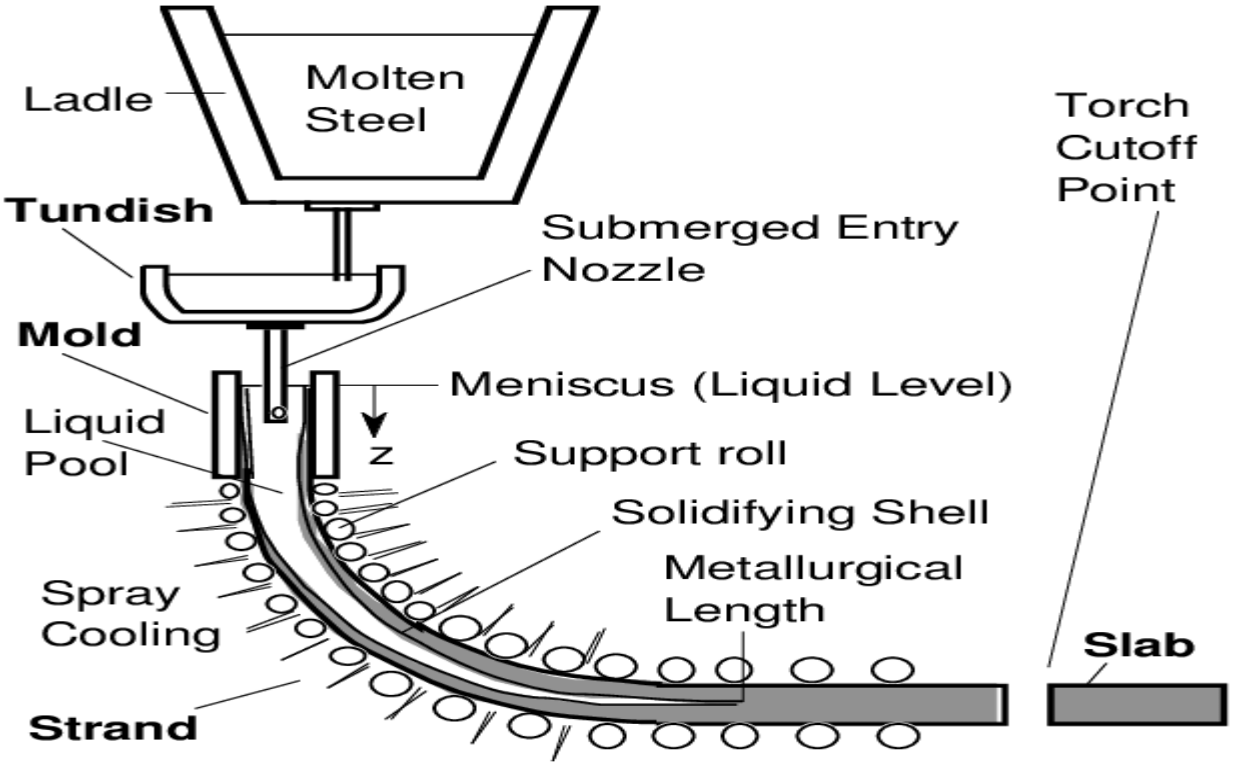


Die casting is a metal casting process that is characterized by forcing molten metal under high pressure into a mold cavity.

Die casting is often used to make components for the automotive industry or decorative hardware and many other small components



Continuous casting, is the process whereby molten metal is solidified into a "semi-finished" billet, bloom, or slab for subsequent rolling in the finishing mills.



Metal Fabrication Methods (vi)



• **Powder metallurgy** is a term covering a wide range of ways in which materials or components are made from metal powders.

Typical Powder Metallurgy parts are made in three basic steps: the blending of the metal powder (pulverization), die compaction, and sintering of the product.

Welding is a fabrication process whereby two or more parts of a metal are fused together by means of heat or pressure.

Heat Treatment Temperature-Time Paths

a) Full Annealing

b) Quenching

c) Tempering (Tempered Martensite)

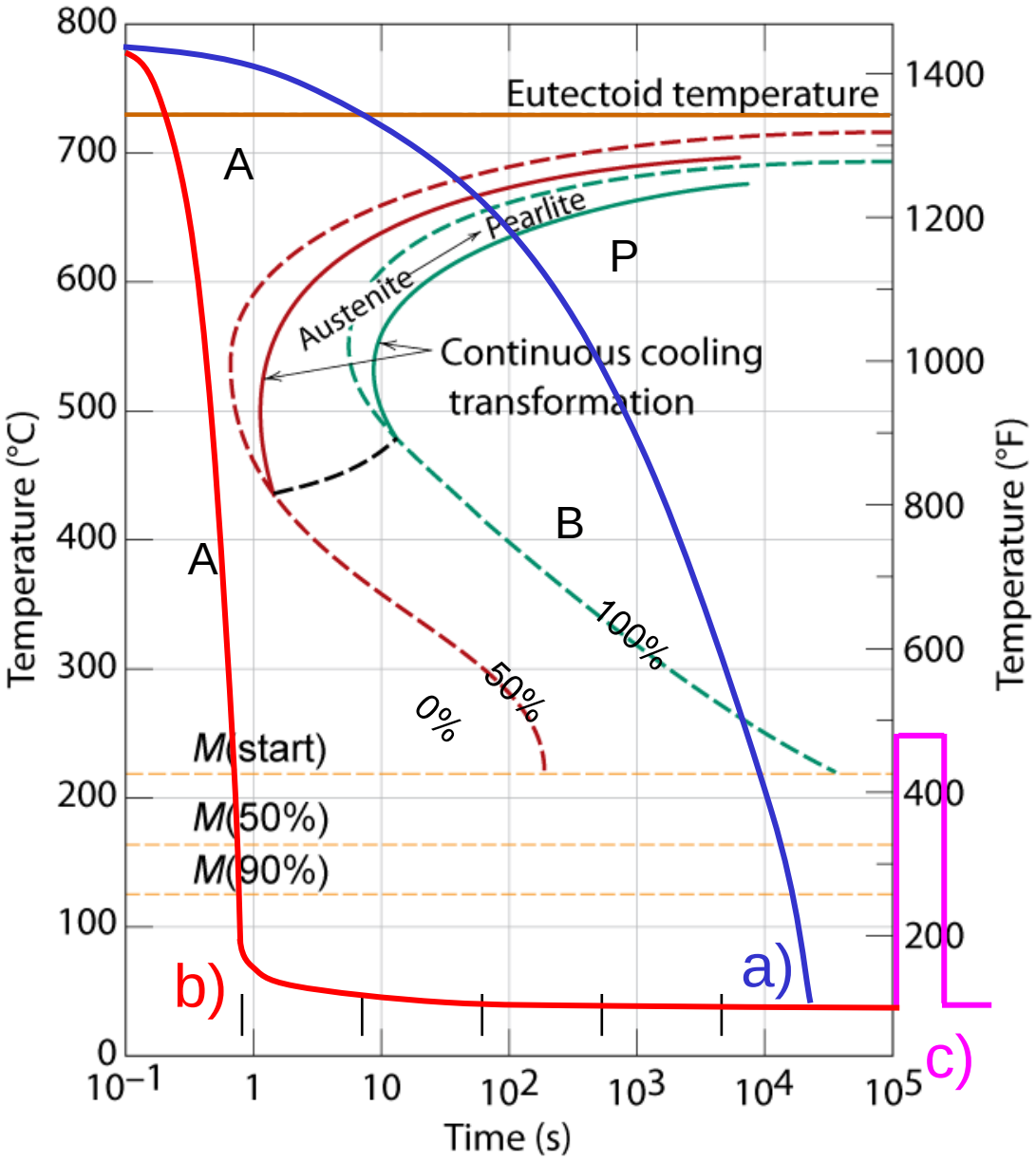


Fig. 12.25, *Callister & Rethwisch 9e*. [Adapted from H. Boyer (Editor), Atlas of Isothermal Transformation and Cooling Transformation Diagrams, 1977. Reproduced by permission of ASM International, Materials Park, OH.]

Thermal Processing of Metals

Annealing is a heat treatment process that changes the physical properties of a material to increase ductility and reduce the hardness to make it more ductile.

Stress Relief Annealing is a heat-treatment procedure of casts with the aim to reduce inner stress within the castings through heating and slow cooling-down.

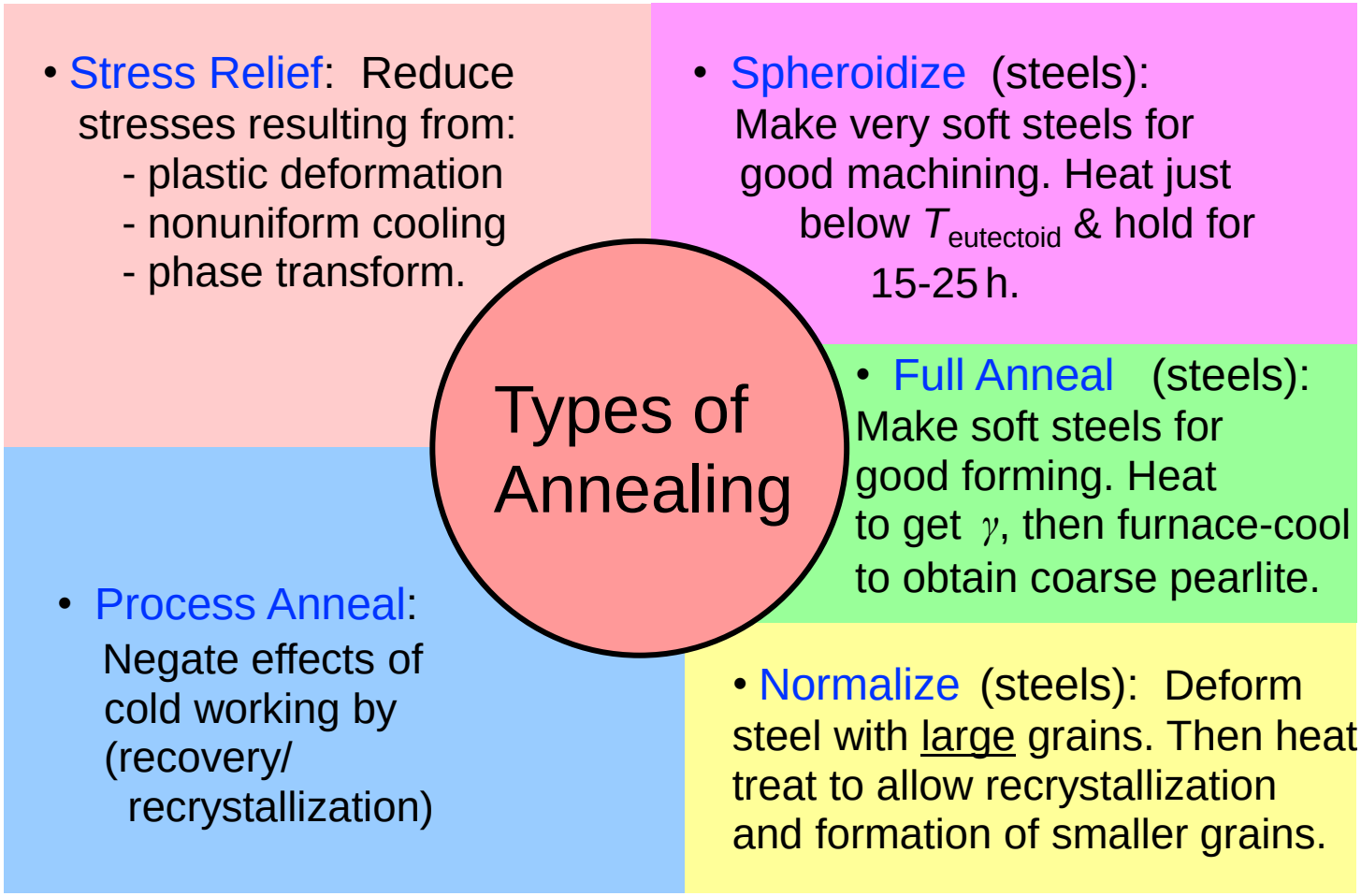
Spheroidize Annealing is a type of annealing process involving controlled heating and cooling to produce a microstructure consisting of spherical (spheroidal) carbides in a ferrite matrix.*

Process Annealing the process of softening steel by heating it to a temperature near but below the transformation (Solid to Liquid) range and then cooling slowly.

Normalizing is a heat treatment process in which steel is heated about its recrystallization temperature and slow cooled in air

Full annealing is the process by which distorted cold-worked lattice structure is changed back to one that is strain-free through the application of heat.

Thermal Processing of Metals



Hardenability -- Steels

- Hardenability – measure of the ability to form martensite
- Jominy end quench test used to measure hardenability.

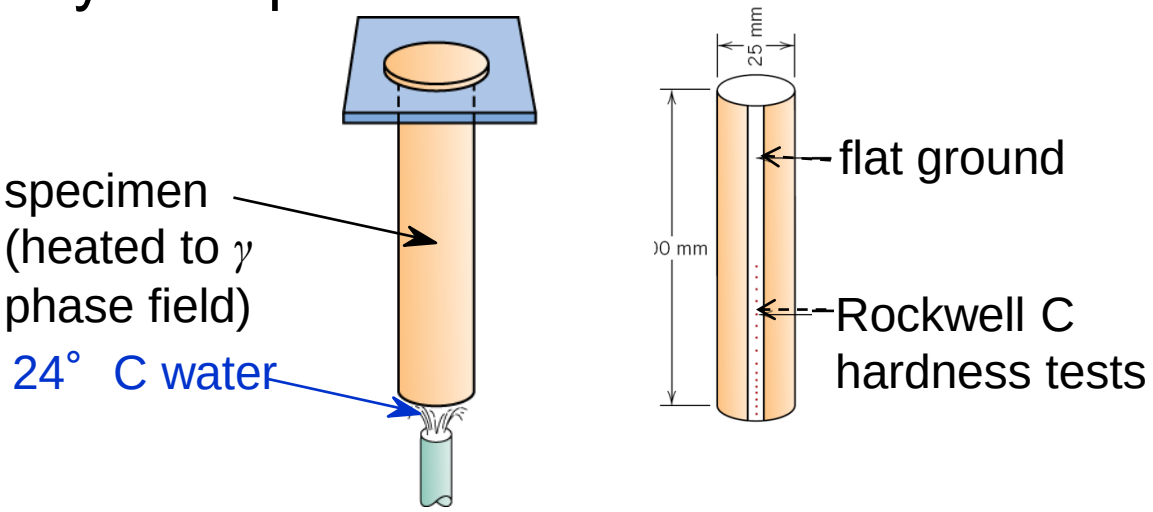


Fig. 17.5, *Callister & Rethwisch 9e*.
(Adapted from A.G. Guy, *Essentials of Materials Science*, McGraw-Hill Book Company, New York, 1978.)

- Plot hardness versus distance from the quenched end.

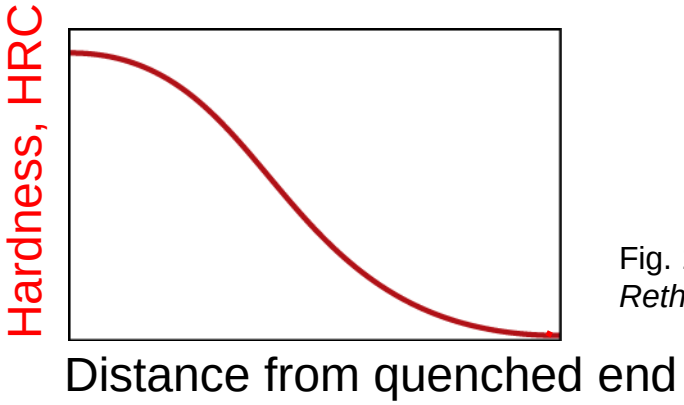


Fig. 17.6, *Callister & Rethwisch 9e*.

Influences of Cooling Medium & Specimen Geometry on Hardness

- Effect of cooling medium:

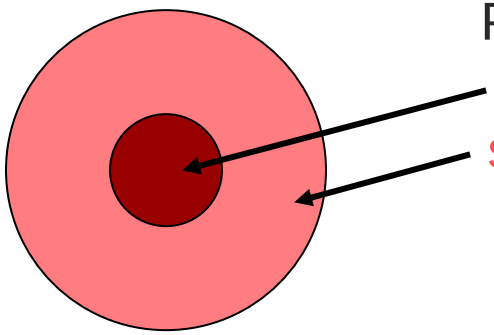
Medium	Severity of cooling	Hardness
air	low	low
oil	moderate	moderate
water	high	high

- Effect of specimen geometry:

  When surface area-to-volume ratio increases:

  -- cooling rate throughout interior increases

  -- hardness throughout interior increases



Position	Cooling rate	Hardness
center	low	low
surface	high	high

Ceramic Fabrication Methods (i)

GLASS FORMING

PARTICULATE FORMING

CEMENTATION

Fiber drawing is a process in which a comparatively thicker glass structure is heated and drawn into extended lengths of micro-structured fiber glass.

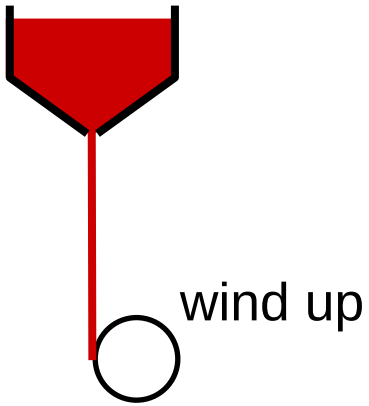
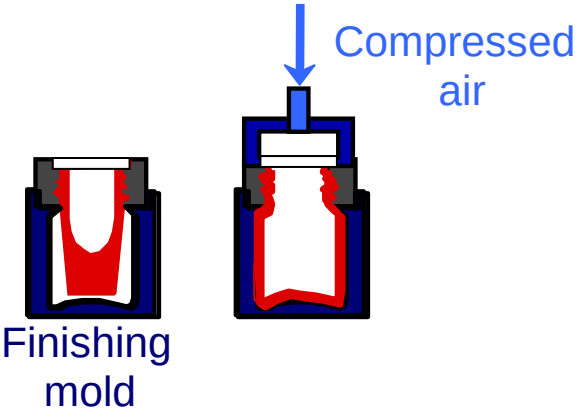
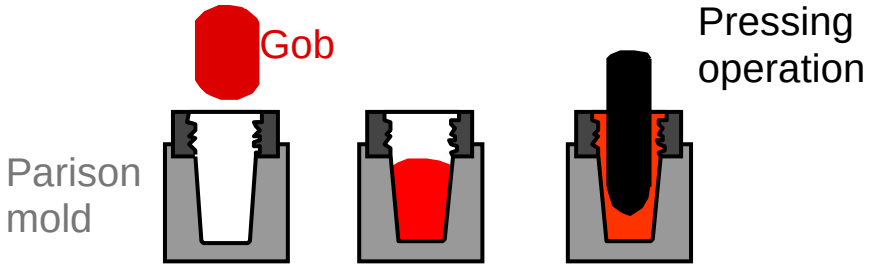


Fig. 17.25, *Callister & Rethwisch 9e*. (Adapted from C.J. Phillips, *Glass: The Miracle Maker*. Reproduced by permission of Pittman Publishing Ltd., London.)

Glassblowing is a glass forming technique that involves inflating molten glass into a bubble (or parison) with the aid of a blow pipe (or blow tube).



Glass Pressing is a forming or shaping of glass by pressing molten glass into a mold using a plunger



Sheet Glass Forming

- Sheet forming – continuous casting
  - sheets are formed by floating the molten glass on a pool of molten tin

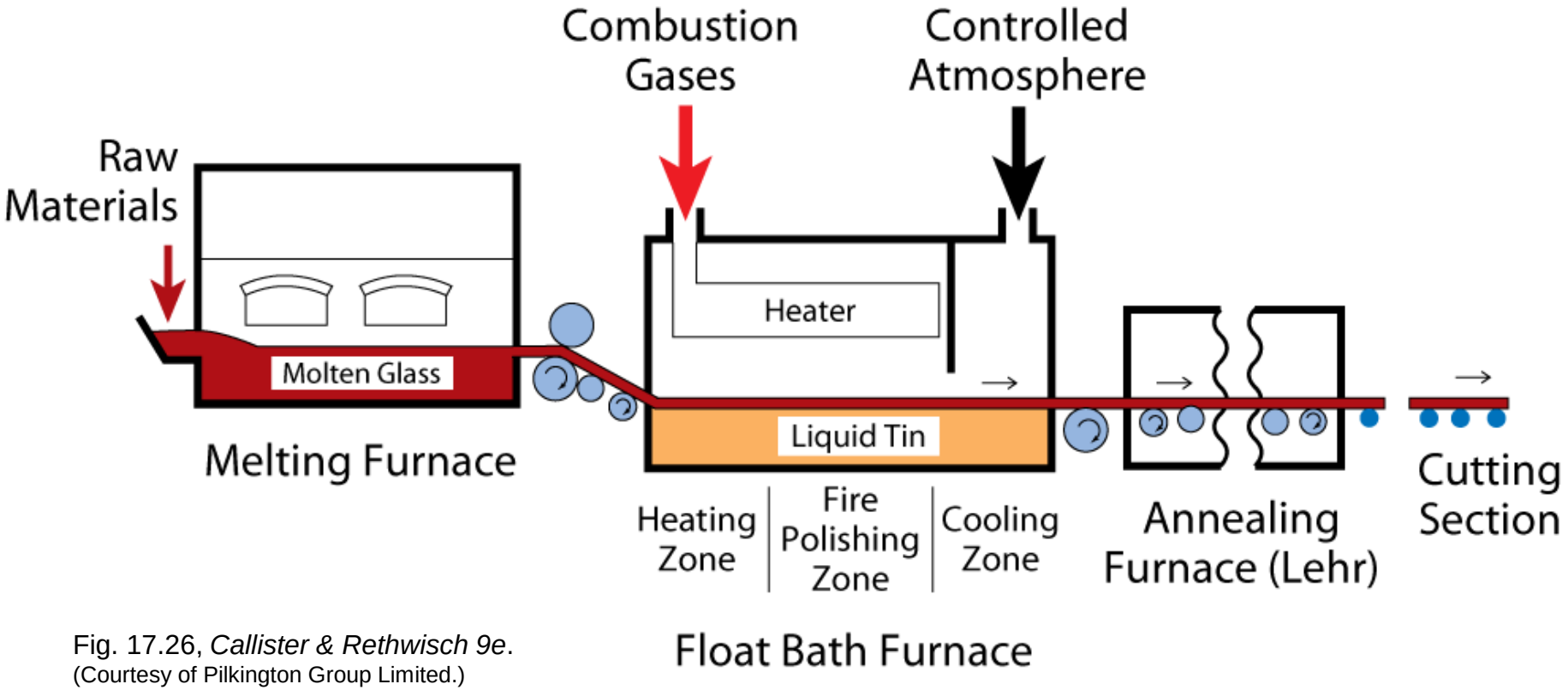


Fig. 17.26, *Callister & Rethwisch 9e*.
(Courtesy of Pilkington Group Limited.)

Heat Treating Glass

- **Annealing:**
  -- removes internal stresses caused by uneven cooling

- **Tempering:** heating and then quickly cool it down to make it stronger and more durable than annealed or untreated glass.

  -- puts surface of glass part into compression
  -- suppresses growth of cracks from surface scratches.

Ceramic Fabrication Methods

(50%) 1. Clay

(25%) 2. Filler – e.g. quartz (finely ground)

(25%) 3. Fluxing agent (Feldspar)

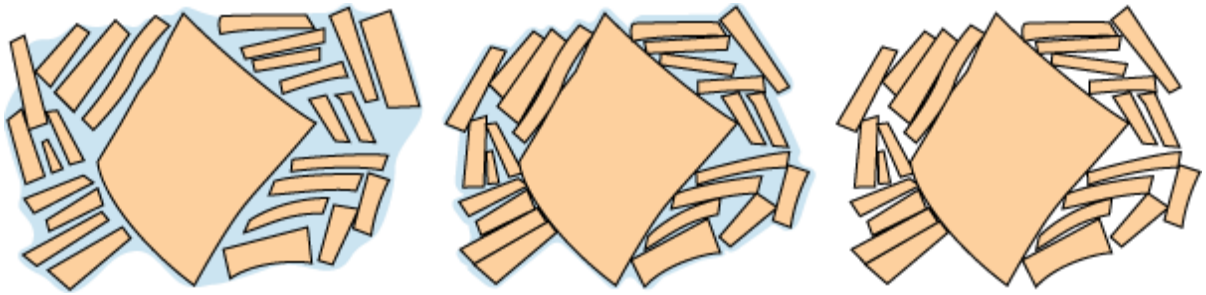
Hydroplasticity of Clay

- Clay is inexpensive and compresses mainly of aluminosilicate

- When water is added to clay
  -- water molecules fit in between layered sheets
  -- reduces degree of van der Waals bonding
  -- when external forces applied – clay particles free to move past one another – becomes hydroplastic

Drying and Firing

- Drying: as water is removed - interparticle spacings decrease – shrinkage .



wet body partially dry completely dry

Fig. 17.30, *Callister & Rethwisch 9e*. (From W.D. Kingery, *Introduction to Ceramics*, Copyright © 1960 by John Wiley & Sons, New York. Reprinted by permission of John Wiley & Sons, Inc.)

Drying too fast causes sample to warp or crack due to non-uniform shrinkage

- Firing:
  -- heat treatment between 900-1400˚ C
  -- vitrification: liquid glass forms from clay and flux – flows between SiO_2 particles. (Flux lowers melting temperature).

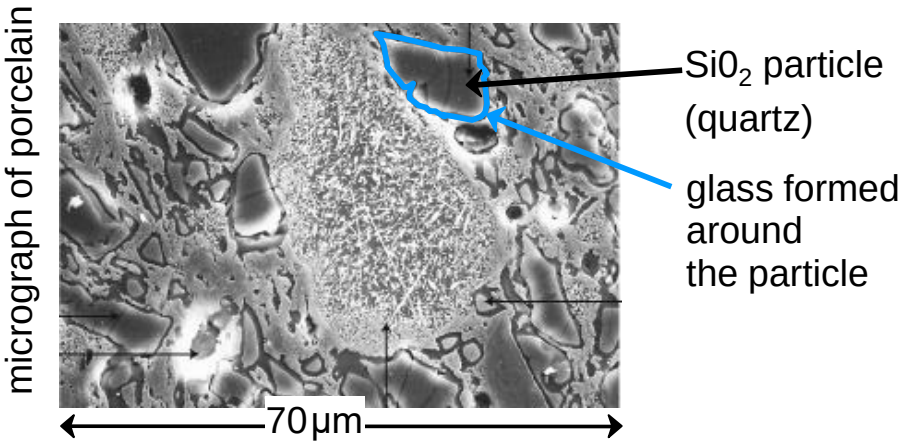
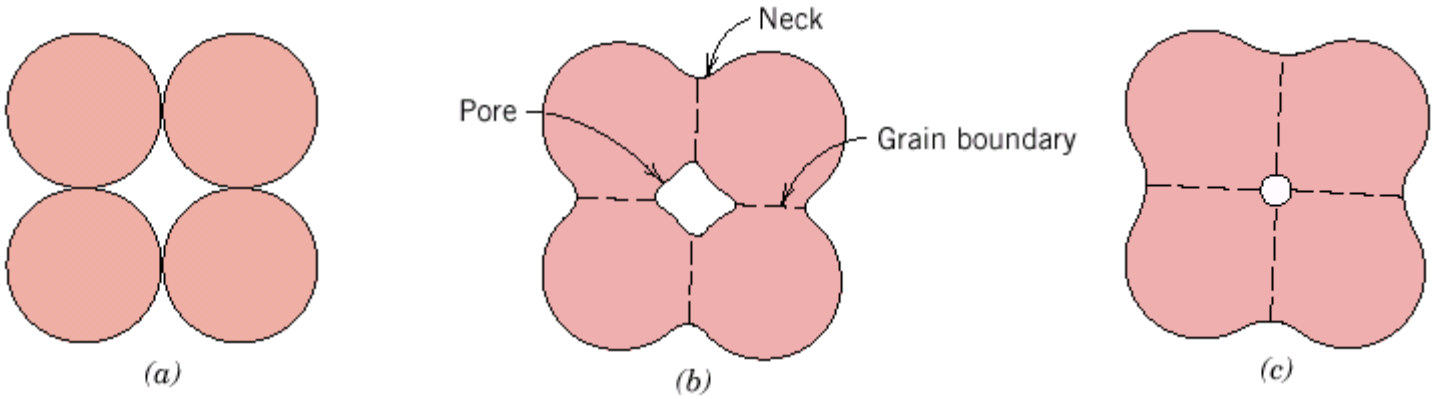


Fig. 17.31, *Callister & Rethwisch 9e*. (Courtesy H.G. Brinkies, Swinburne University of Technology, Hawthorn Campus, Hawthorn, Victoria, Australia.)

Sintering

Sintering occurs during firing of a piece that has been powder pressed

-- powder particles coalesce and reduction of pore size



Polymer Formation

- **Polymer** is a large molecule composed of repeating structural units connected by covalent chemical bonds.

- **Polymerization,** any process in which relatively small molecules, called monomers, combine chemically to produce a very large chainlike or network molecule, called a polymer.

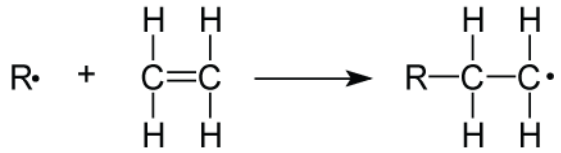
- There are two types of polymerization
  - Addition (or chain) polymerization
  - Condensation (step) polymerization

In **Condensation Polymerization**, each step of the process is accompanied by the formation of a molecule of some simple compound, often water.

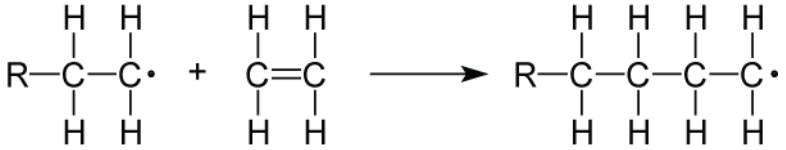
In **Addition Polymerization,** monomers react to form a polymer without the formation of by-products. Addition polymerizations usually are carried out in the presence of catalysts, which in certain cases exert control over structural details that have important effects on the properties of the polymer.

Addition (Chain) Polymerization

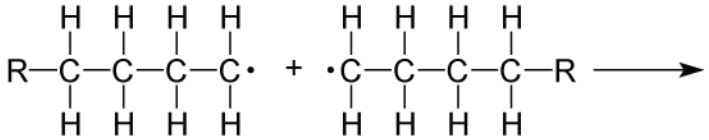
– **Initiation**

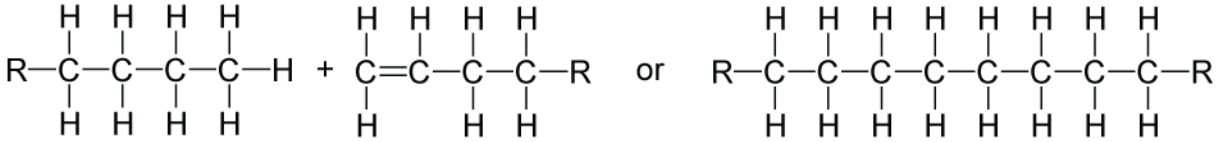


– **Propagation**



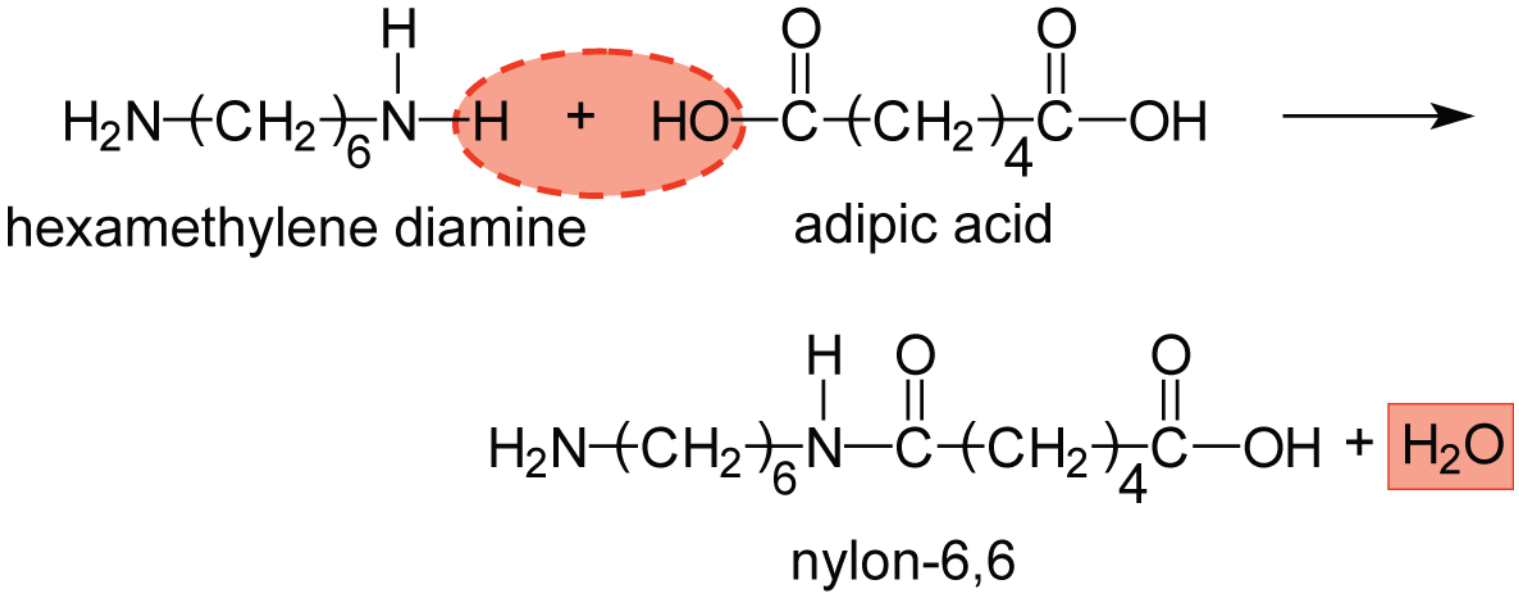
– **Termination**





Disproportionation Combination

Condensation (Step) Polymerization



Polymer Additives

Improve mechanical properties, processability, durability, etc.

- Fillers
  - Added to improve tensile strength & abrasion resistance, toughness & decrease cost
  - ex: carbon black, silica gel, wood flour, glass, limestone, talc, etc.
- Plasticizers
  - Added to reduce the glass transition temperature T_g below room temperature
  - Presence of plasticizer transforms brittle polymer to a ductile one
  - Commonly added to PVC - otherwise it is brittle

Polymer Additives (cont.)

- Stabilizers
  - Antioxidants
  - UV protectants
- Lubricants
  - Added to allow easier processing
  - polymer “slides” through dies easier
  - ex: sodium stearate
- Colorants
  - Dyes and pigments
- Flame Retardants
  - Substances containing chlorine, fluorine, and boron

Processing of Plastics

- Thermoplastic
  - can be reversibly cooled & reheated, i.e. recycled
  - heat until soft, shape as desired, then cool
  - ex: polyethylene, polypropylene, polystyrene.
- Thermoset
  - when heated forms a molecular network (chemical reaction)
  - degrades (doesn't melt) when heated
  - a prepolymer molded into desired shape, then chemical reaction occurs
  - ex: urethane, epoxy

Processing Plastics – Compression Molding

Thermoplastics and thermosets

- polymer and additives placed in mold cavity
- mold heated and pressure applied
- fluid polymer assumes shape of mold

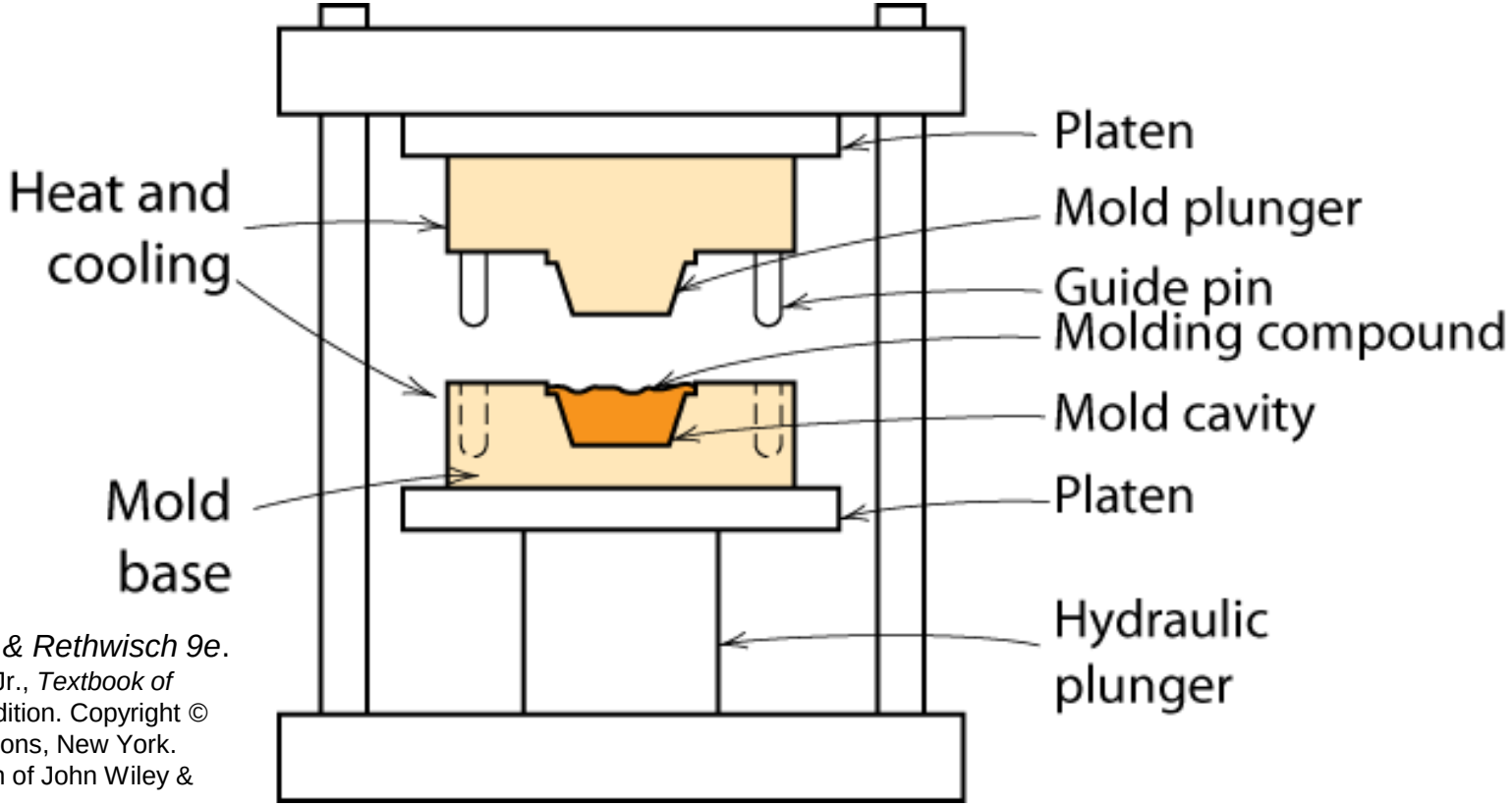


Fig. 17.36, *Callister & Rethwisch 9e*. (From F. W. Billmeyer, Jr., *Textbook of Polymer Science*, 3rd edition. Copyright © 1984 by John Wiley & Sons, New York. Reprinted by permission of John Wiley & Sons, Inc.)

Processing Plastics – Injection Molding

Thermoplastics and some thermosets

- when ram retracts, plastic pellets drop from hopper into barrel
- ram forces plastic into the heating chamber (around the spreader) where the plastic melts as it moves forward
- molten plastic is forced under pressure (injected) into the mold cavity where it assumes the shape of the mold

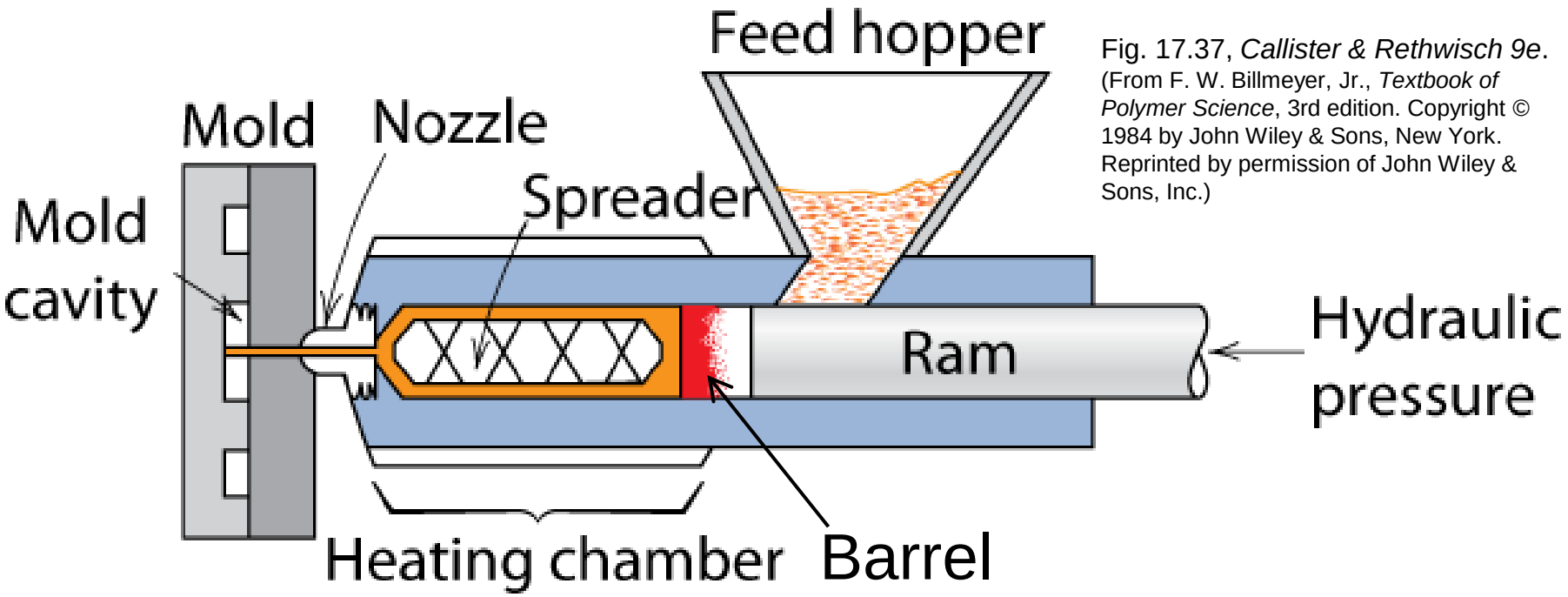


Fig. 17.37, *Callister & Rethwisch 9e*. (From F. W. Billmeyer, Jr., *Textbook of Polymer Science*, 3rd edition. Copyright © 1984 by John Wiley & Sons, New York. Reprinted by permission of John Wiley & Sons, Inc.)

Processing Plastics – Extrusion

thermoplastics

- plastic pellets drop from hopper onto the turning screw
- plastic pellets melt as the turning screw pushes them forward by the heaters
- molten polymer is forced under pressure through the shaping die to form the final product (extrudate)

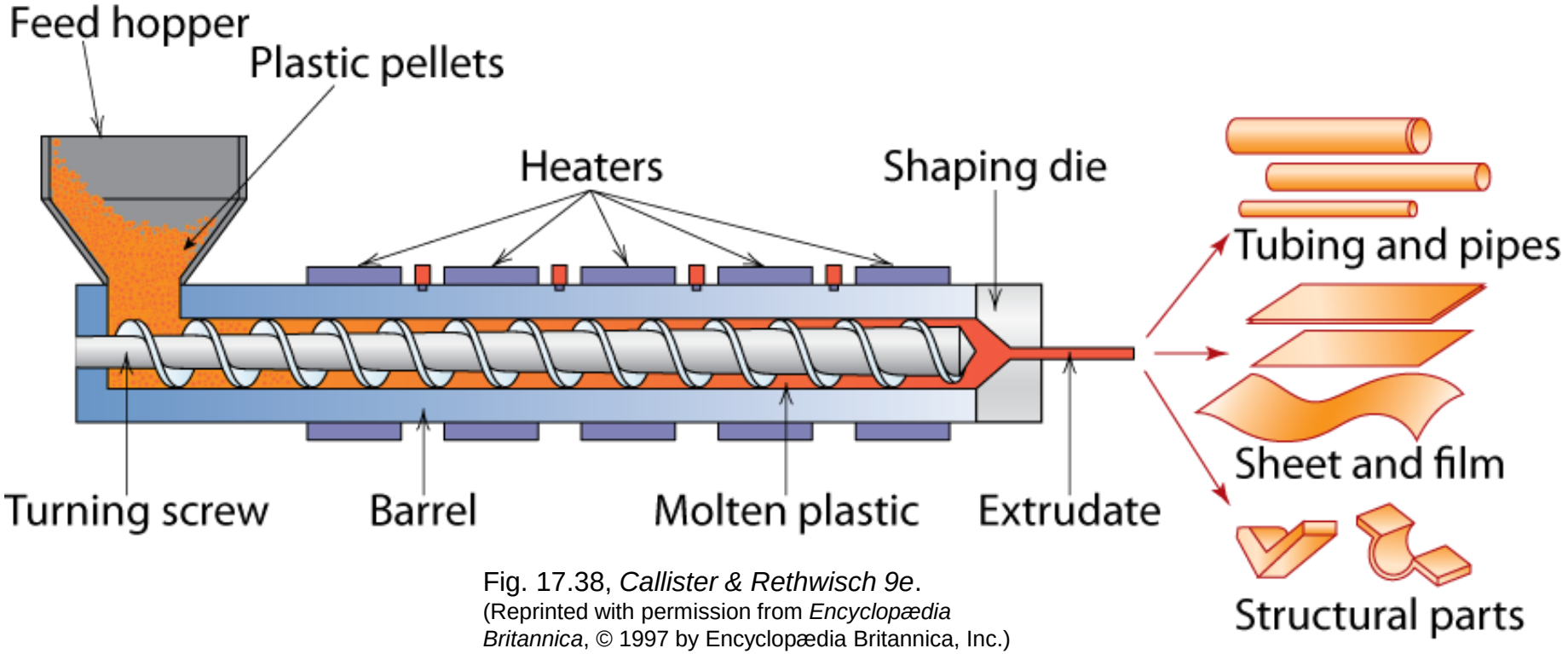


Fig. 17.38, *Callister & Rethwisch 9e*.
(Reprinted with permission from *Encyclopædia Britannica*, © 1997 by Encyclopædia Britannica, Inc.)

Summary

- Metal fabrication techniques:
  -- forming, casting, miscellaneous.
- Hardenability of metals
  -- measure of ability of a steel to be heat treated.
  -- increases with alloy content.
- Precipitation hardening
  --hardening, strengthening due to formation of precipitate particles.
  --Al, Mg alloys precipitation hardenable.

Summary (Cont.)

- Ceramic Fabrication techniques:
  -- glass forming (pressing, blowing, fiber drawing).
  -- particulate forming (hydroplastic forming, slip casting, powder pressing, tape casting)
  -- cementation
- Heat treating procedures for ceramics
  -- glasses—annealing, tempering
  -- particulate formed pieces—drying, firing (sintering)
- Polymer Processing
  -- compression and injection molding, extrusion, blown film extrusion
- Polymer melting and glass transition temperatures

ANNOUNCEMENTS

Reading:

Core Problems:

Self-help Problems: